

CPU, ALU, Control Unit & Registers — Detailed Lecture Notes

JKSSB Computer Awareness

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Video lesson: [CPU Explained: ALU, Control Unit & Registers](#)

05.1 The CPU and its functional units

The Central Processing Unit (CPU) is the part of the computer that executes instructions. It repeatedly fetches an instruction from main memory, decodes it, and then executes it. To do this work the CPU is organised into three functional units: the Arithmetic Logic Unit (ALU), the Control Unit (CU), and the registers. Data and control signals move between these units over internal buses. The CPU works directly with main memory (RAM); it does not reach into secondary storage such as a hard disk by itself.

Definition (CPU (Central Processing Unit)). *The processor that fetches, decodes, and executes instructions, made up of the ALU, the Control Unit, and registers.*

A simple way to keep the roles apart is this: the ALU computes, the Control Unit coordinates, and the registers hold the working values. Almost every objective question about this topic turns on not swapping those three roles.

05.2 The Arithmetic Logic Unit (ALU)

The ALU is the computing part of the CPU. It performs two broad families of operations. **Arithmetic** operations include addition, subtraction, increment, decrement, multiplication, and division. **Logical** operations include AND, OR, NOT, XOR, shifts, and comparisons. The ALU operates on data held in binary form, and its operations are carried out with logic-gate operations on those binary values. When the ALU finishes, it returns its result to a register or the accumulator; it does not keep a permanent copy.

Definition (ALU (Arithmetic Logic Unit)). *The CPU unit that performs arithmetic and logical operations on binary data using logic gates.*

Example 05.2.1. *For inputs 1 and 1, an addition produces the binary result for two, and a logical XOR produces 0 because the two bits are equal. Both results are produced by the same unit — the ALU — and handed back for further use.*

05.3 The Control Unit (CU)

The Control Unit directs and sequences the operations of the computer. It decodes each instruction and then issues the control signals that time and trigger the other units. It also controls the flow of data between the ALU, the registers, and memory. What the Control Unit does **not** do is

compute: it does not add, subtract, or carry out logical operations. Those are the ALU's job. Students frequently write "the Control Unit performs the calculations," which is exactly the error to avoid.

Definition (Control Unit (CU)). *The CPU unit that directs and sequences operations, decodes instructions, and issues control signals; it does not perform arithmetic or logic.*

Example 05.3.1. *When an addition instruction is executed, the Control Unit fetches and decodes the instruction and signals the ALU to add the two values. The ALU performs the addition; the Control Unit only directs the sequence.*

05.4 Registers

Registers are the smallest and fastest storage in a computer. They are located inside the CPU, so they are faster than cache and far faster than main memory. Because they are very small, they hold only the values needed for the work currently in progress: instructions, addresses, data, and intermediate results. The following table lists the key registers and their purposes.

Register	Purpose
Program Counter (PC / IP)	Holds the address of the next instruction to be fetched.
Instruction Register (IR)	Holds the instruction currently being decoded or executed.
Memory Address Register (MAR)	Holds the address of the memory location to be accessed.
Memory Data Register (MDR)	Holds the data being transferred to or from memory.
Accumulator (ACC)	Holds intermediate arithmetic and logic results.
General-purpose registers	Hold temporary operands and results.
Status / flags register	Holds condition flags such as zero, carry, sign, and overflow.

The Program Counter is the register most often confused. It holds the address of the **next** instruction to fetch, not the address of the instruction currently being executed. The instruction currently being decoded or executed is held in the Instruction Register. Memory access is split between two registers: the MAR carries the address, and the MDR carries the data itself. The accumulator is the natural home for an ALU result that will be used again immediately.

Example 05.4.1. *Suppose the CPU is about to fetch the instruction at address 200. The Program Counter holds 200. Once the instruction is fetched, it is placed in the Instruction Register for decoding. If the instruction needs to read a value stored at address 900, the MAR holds 900 and the MDR holds the value brought back from memory.*

05.5 Clock speed, cycles, cores, and threads

The CPU's operations are paced by a clock. **Clock speed** is measured in hertz (Hz); one gigahertz (GHz) is one billion clock cycles per second. A **clock cycle** is a single tick of that clock. Do not assume every instruction takes exactly one cycle: different instructions can require different numbers of clock cycles. A **core** is an independent processing unit within the CPU; a quad-core chip contains four cores. A **thread** is a stream of instructions, and a system can run multiple threads across the available cores.

Definition (Clock cycle). *One tick of the system clock; the basic unit of timing for CPU operations.*

Example 05.5.1. *The office desktop in our running example is a quad-core machine running at 2.4 GHz. It has four independent cores, and each core can complete clock cycles at about 2.4 billion per second. Running a spreadsheet calculation and a browser at the same time lets different threads keep the cores busy.*

05.6 Exam retrieval and common confusions

Three distinctions prevent most errors on this topic.

1. Do not swap the ALU and the Control Unit. The ALU performs arithmetic and logic; the Control Unit only directs operations and issues control signals.
2. Do not say the Program Counter holds the current instruction. It holds the address of the **next** instruction; the current instruction is in the Instruction Register.
3. Do not place registers below cache or RAM in speed. Registers are the fastest storage, and the accumulator is where intermediate arithmetic and logic results are kept.

Exercises

1. State the three main functional units of the CPU and one role of each.
2. Expand ALU, CU, PC, IR, MAR, and MDR.
3. Does the Control Unit perform arithmetic? Explain in one line.
4. Which register holds the address of the next instruction, and which register holds the instruction currently being executed?
5. Why are registers described as the fastest storage, and what does the accumulator hold?

Solutions

1. ALU — performs arithmetic and logical operations on binary data; Control Unit — directs and sequences operations and issues control signals; registers — hold instructions, addresses, data, and intermediate results.
2. ALU = Arithmetic Logic Unit; CU = Control Unit; PC = Program Counter; IR = Instruction Register; MAR = Memory Address Register; MDR = Memory Data Register.
3. No. The Control Unit directs and coordinates operations; the ALU performs the arithmetic and logical calculations.
4. The Program Counter (PC/IP) holds the address of the next instruction; the Instruction Register (IR) holds the instruction currently being decoded or executed.
5. Registers are located inside the CPU and are smaller and faster than cache and RAM. The accumulator (ACC) holds intermediate arithmetic and logic results.